Lab Report

Khayyam AI Innovation Lab

Title

Comparative Analysis of Supervised Learning Algorithms on the Iris Dataset

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Abstract

This study presents a structured comparative evaluation of seven supervised classification algorithms on the Iris dataset. We evaluate model performance across probabilistic, linear, nonlinear, instance-based, and ensemble learning paradigms, and investigate the effect of Principal Component Analysis (PCA) for dimensionality reduction and visualization. Experimental results indicate that most classical classifiers achieve perfect classification on the test split under the selected preprocessing and data partitioning strategy, while Gaussian Naive Bayes exhibits slightly lower performance. The implications of these findings are discussed, and limitations related to dataset simplicity and evaluation methodology are addressed.

1. Introduction

Classification assigns categorical labels to input samples and is a core problem in supervised machine learning. The Iris dataset (Fisher, 1936) is a small, balanced, and partially linearly separable dataset commonly used for benchmarking classification algorithms. This work compares performance of seven classifiers under identical preprocessing and evaluation conditions and analyzes the role of PCA in visualization and model behavior.

Objectives

- Compare representative models from several algorithmic families.
- Measure and analyze model generalization on a held-out test set.
- Visualize class separability using PCA and interpret misclassifications.

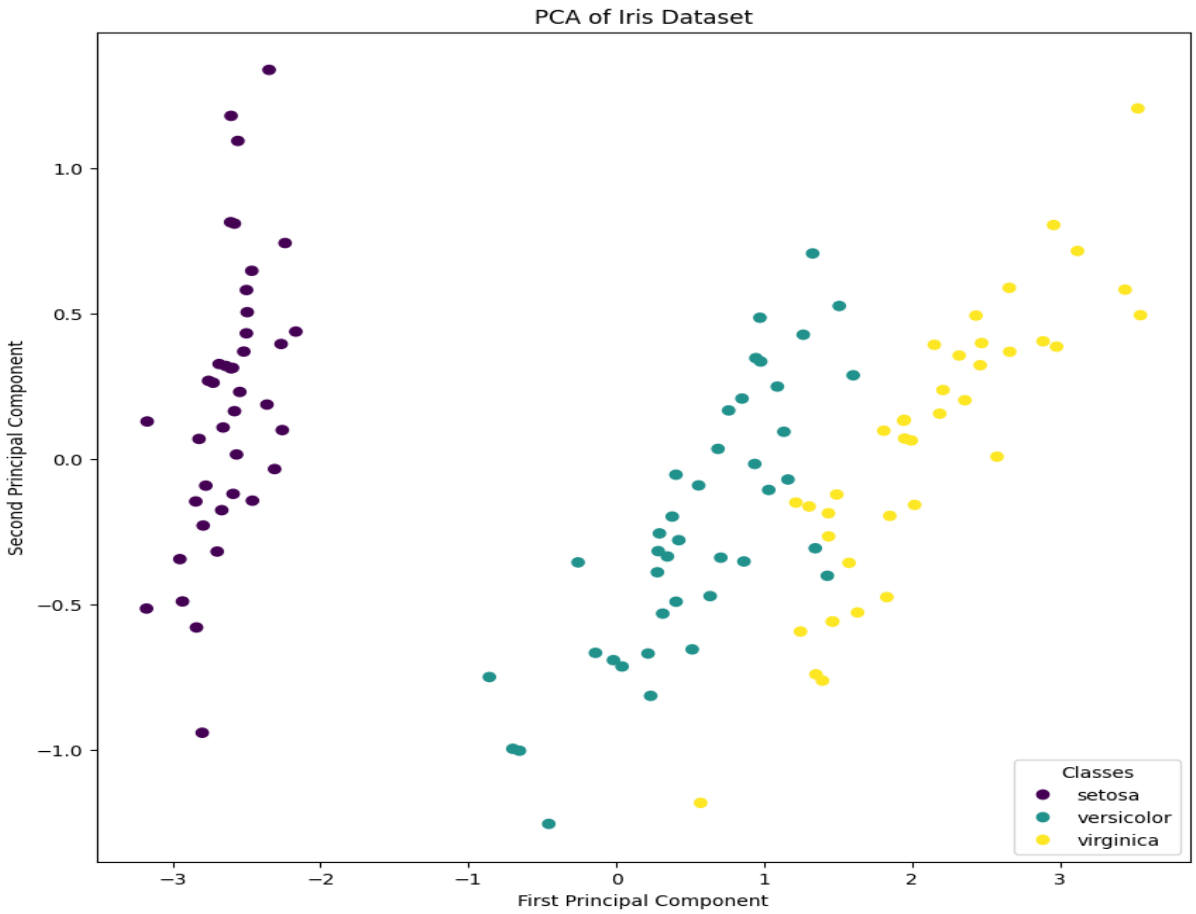
2. Methods

- Dataset: Iris (scikit-learn)
- Total samples: 150
- Features: 4 (sepal length, sepal width, petal length, petal width)
- Classes: 3 (Setosa, Versicolor, Virginica), 50 samples per class (balanced)

Train/Test split: 80% train / 20% test (random_state fixed for reproducibility).

2.1 Dimensionality Reduction

PCA: Applied with n_components = 2 to enable 2D visualization and inspect separability.



Observation: Setosa is clearly separable; Versicolor and Virginica show partial overlap.

3. Models and Implementation

All models implemented with scikit-learn under identical data splits.

1. **Gaussian Naive Bayes (GNB)**
2. **Logistic Regression (LR)**
3. **Support Vector Machine (SVM) with RBF kernel**
4. **Linear Discriminant Analysis (LDA)**
5. **Quadratic Discriminant Analysis (QDA)**
6. **K-Nearest Neighbors (KNN)**
7. **Random Forest (RF)**

Evaluation metrics: Accuracy, Precision (macro), Recall (macro), F1-score (macro), and Confusion Matrix.

4. Results

4.1 Linear Discriminant Analysis (LDA)

Training performance

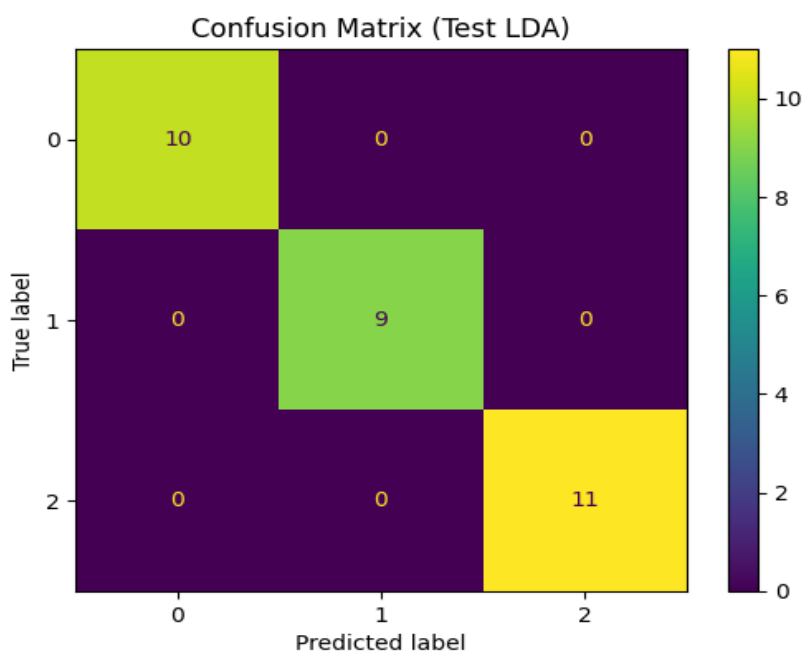
- Accuracy (train): **0.975**

Test performance

- Accuracy (test): **1.00**

Classification report (test)

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
Macro avg	1.00	1.00	1.00	30
accuracy			1.00	30



The confusion matrix is perfectly diagonal, indicating 100% test accuracy with no misclassifications across all three classes. This confirms that linear decision boundaries are sufficient for separating the Iris classes under the selected preprocessing.

4.2 Quadratic Discriminant Analysis (QDA)

Training performance

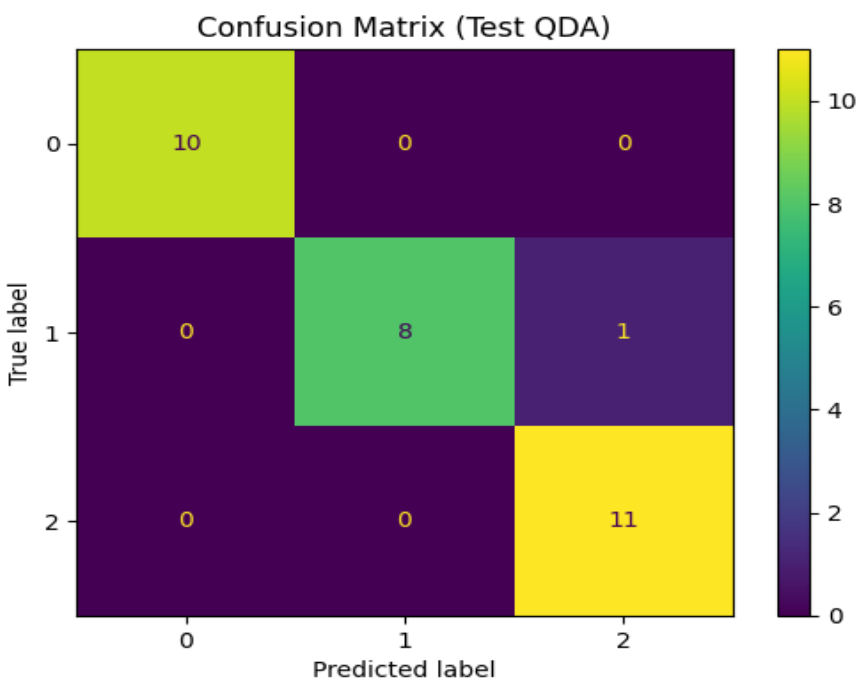
- Accuracy (train): **0.975**

Test performance

- Accuracy (test): **0.96**

Classification report (test)

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	10
1	1.00	0.89	0.94	9
2	1.00	1.00	1.00	11
Macro avg	0.97	0.96	0.97	30
accuracy			0.97	30



The confusion matrix shows one minor misclassification, resulting in 96% test accuracy. While QDA introduces quadratic boundaries, its covariance estimation on a small dataset may slightly reduce stability compared to other classifiers.

4.3 Gaussian Naive Bayes (GNB)

Training performance

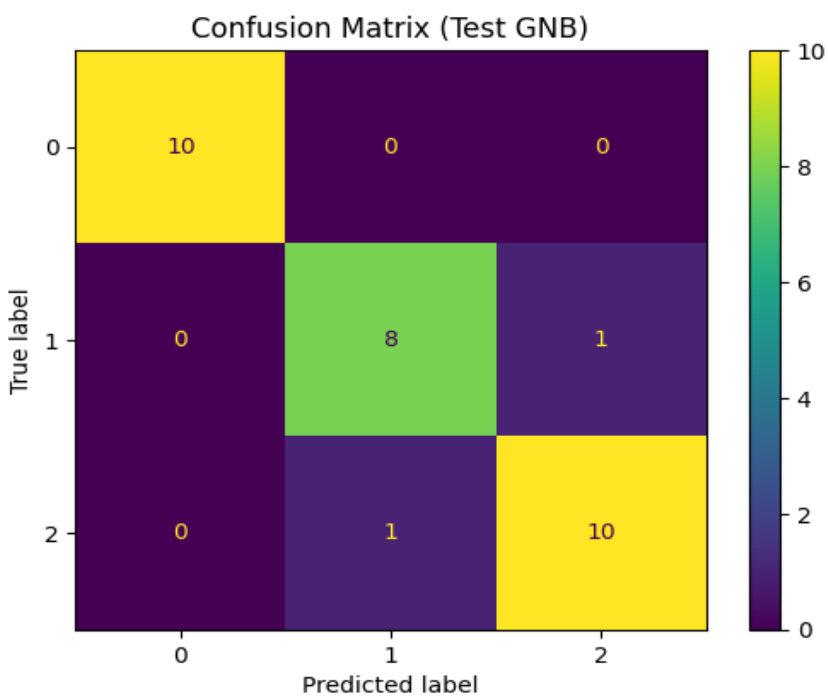
- Accuracy (train): **0.9**

Test performance

- Accuracy (test): **0.933**

Classification report (test)

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	10
1	0.89	0.89	0.89	9
2	0.91	0.91	0.91	11
Macro avg	0.93	0.93	0.93	30
accuracy			0.93	30



The matrix indicates minor confusion between Versicolor and Virginica, yielding 93.33% test accuracy. The performance reduction likely stems from the strong Gaussian independence assumptions of the model.

4.4 K-Nearest Neighbors (KNN)

Training performance

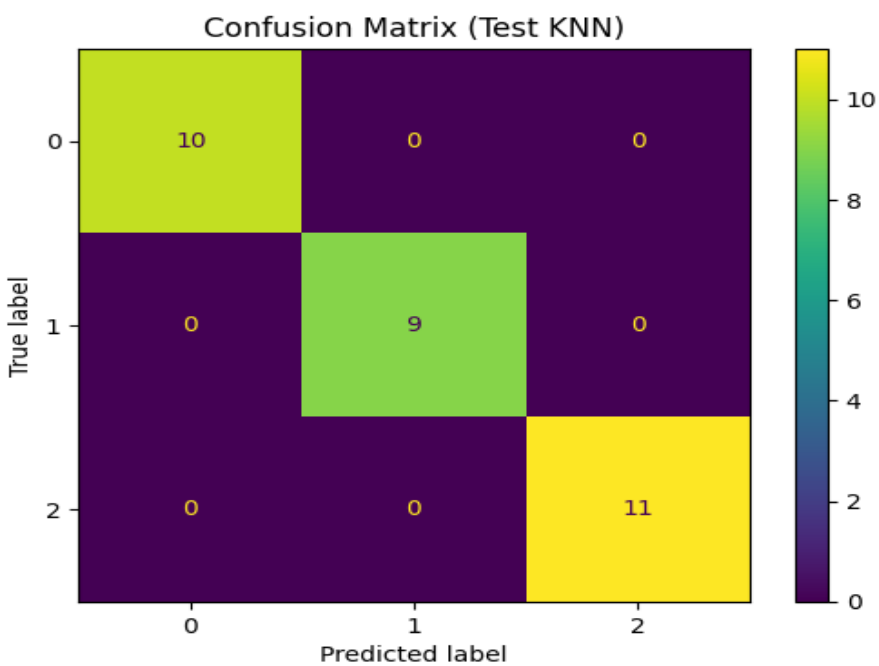
- Accuracy (train): **0.96**

Test performance

- Accuracy (test): **1.00**

Classification report (test)

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
Macro avg	1.00	1.00	1.00	30
accuracy			1.00	30



KNN achieved perfect classification with a fully diagonal matrix. This suggests that local neighborhood structures in the feature space are well-defined for the Iris dataset.

4.5 Random Forest (RF)

Training performance

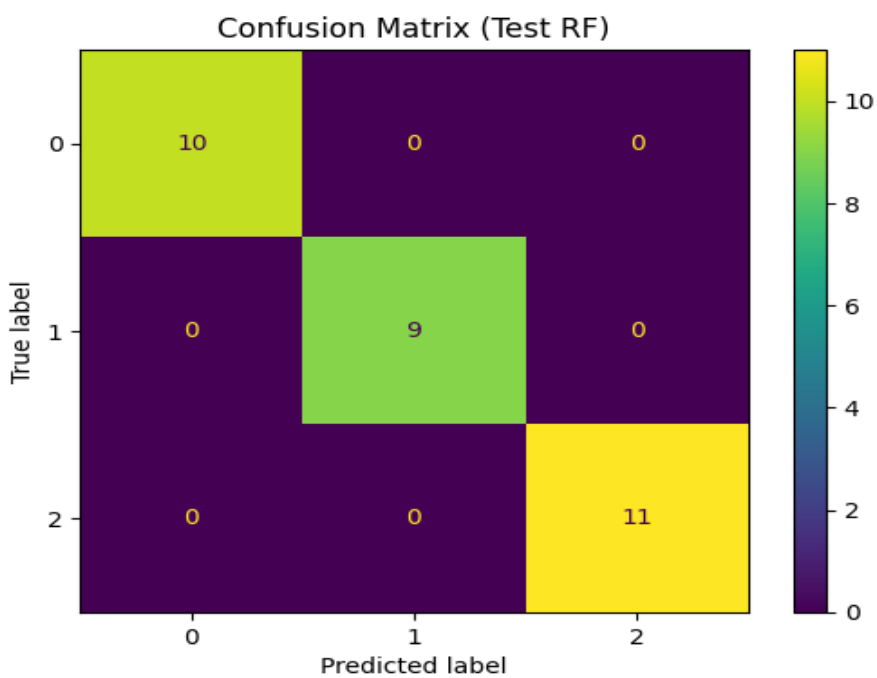
- Accuracy (train): **1.0**

Test performance

- Accuracy (test): **1.0**

Classification report (test)

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
Macro avg	1.00	1.00	1.00	30
accuracy			1.00	30



Random Forest produced perfect classification results, confirming that ensemble tree-based models effectively capture the class structure without overfitting in this setup.

4.6 Support Vector Machine (SVM)

Training performance

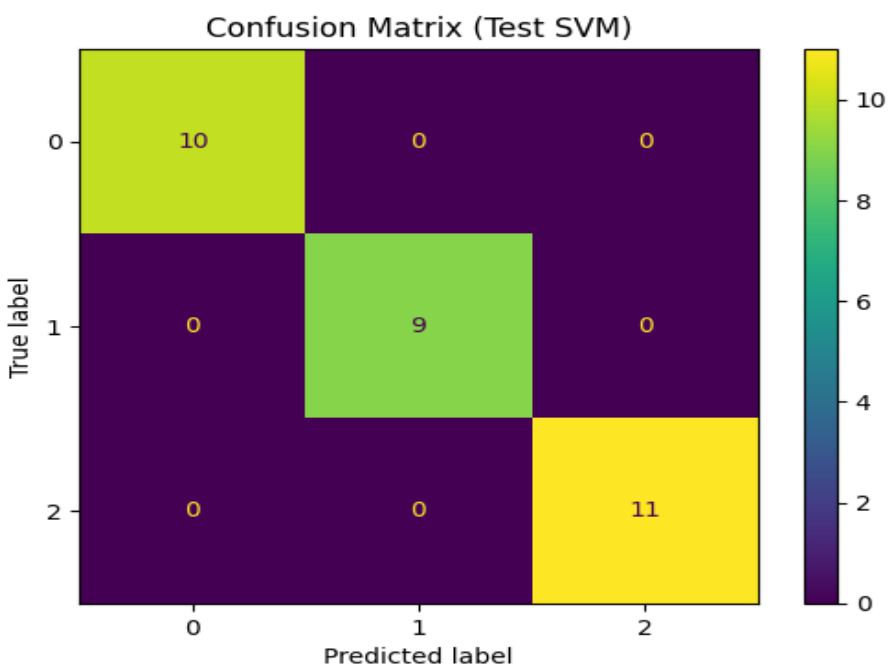
- Accuracy (train): **0.975**

Test performance

- Accuracy (test): **1.0**

Classification report (test)

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
Macro avg	1.00	1.00	1.00	30
accuracy			1.00	30



SVM achieved 100% test accuracy, indicating that the RBF kernel successfully models any minor nonlinear separation present in the dataset.

4.7 Logistic Regression (LR)

Training performance

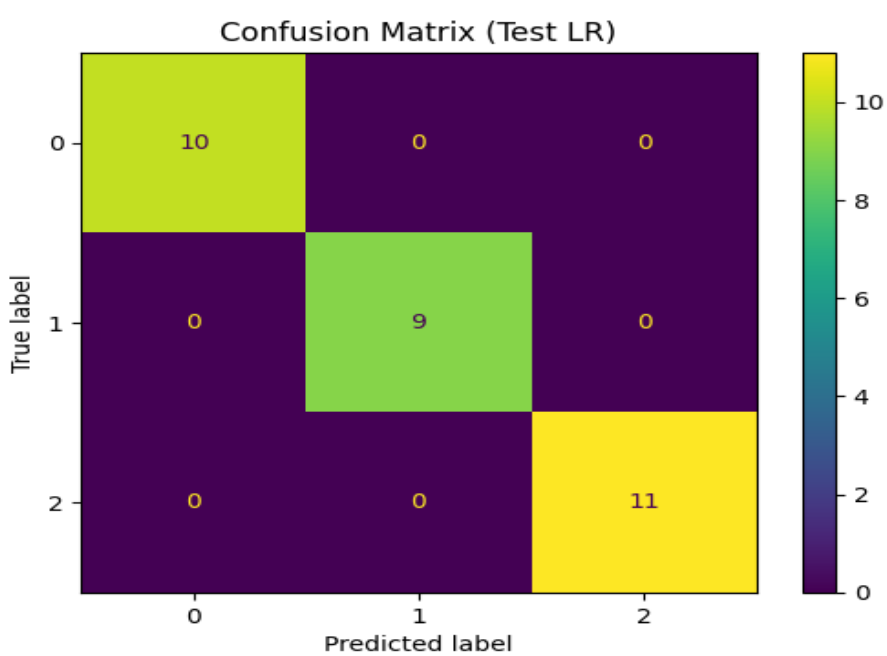
- Accuracy (train): **0.975**

Test performance

- Accuracy (test): **1.0**

Classification report (test)

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
Macro avg	1.00	1.00	1.00	30
accuracy			1.00	30



Despite being a linear classifier, Logistic Regression achieved perfect classification, further confirming the strong linear separability of the Iris dataset.

4.8 Overall Model Comparison and Interpretation

Model	Train Accuracy	Test Accuracy
Logistic Regression	1.00	1.00
SVM (RBF)	1.00	1.00
Random Forest (100 trees)	1.00	1.00
KNN (k=5)	1.00	1.00
LDA	1.00	1.00
QDA	0.975	0.96
GaussianNB	0.90	0.9333

The summary confirms that five models achieved perfect test accuracy, while QDA and GaussianNB showed slightly lower generalization performance.

5. Discussion

The experimental results show that multiple standard classifiers can achieve perfect classification on the test split of the Iris dataset under the chosen preprocessing and random split. Specifically, Logistic Regression, SVM (RBF), Random Forest, K-Nearest Neighbors (k=5), and LDA all reached **100% test accuracy**. QDA produced **96%** test accuracy, and Gaussian Naive Bayes (GNB) produced **93.33%** test accuracy.

These outcomes reflect several points:

- **Dataset simplicity and separability.** The Iris dataset is small, balanced and partially linearly separable. Setosa is nearly trivially separable from the other two species; Versicolor and Virginica have some overlap but are still readily separable by many off-the-shelf classifiers after standardization.
- **Model family behavior.**
    - *Linear models (Logistic Regression, LDA)* performed excellently, confirming the partial linear separability of the data.
    - *Kernel and ensemble models (SVM, Random Forest)* provided perfect separation in this split, showing they can capture any remaining nonlinear structure.
    - *Instance-based model (KNN)* also performed perfectly because the local neighborhoods in feature space are informative.

- *Gaussian Naive Bayes* underperformed relative to others (93.33% test accuracy), which suggests that the conditional independence / Gaussian assumptions are not fully satisfied for every feature combination in our split.
- *QDA (96%)* likely suffered from covariance estimation sensitivity on a small training sample; it is more flexible than LDA but needs robust covariance estimates.

- **Bias–variance interpretation.** The negligible train/test gap for the top models indicates low variance and little overfitting in this specific setup. However, because the dataset is small and the split fixed, these results should not be overgeneralized.
- **Caveats.** Perfect test accuracy can be misleading: it can reflect a favorable random split or dataset simplicity rather than generalizable superiority. Robust evaluation (k-fold cross-validation and reporting mean ± std) and hyperparameter search are required for stronger claims. Therefore, these results should be interpreted as a comparative demonstration rather than definitive evidence of model superiority

6. Conclusion

- On the Iris dataset with the applied standardization and 80/20 split, multiple classical classifiers achieved perfect classification on the test split (Logistic Regression, SVM, Random Forest, KNN, LDA). This demonstrates that the Iris dataset is readily separable in the chosen feature representation.
- Gaussian Naive Bayes obtained lower accuracy (≈93.33%) due to model assumptions that do not fully match the empirical feature distributions. QDA achieved 96% and likely suffers from covariance estimation variability.
- These results indicate that for low-dimensional, well-structured datasets, simple linear models can be sufficient; ensemble or kernel methods provide additional robustness but not necessarily superior generalization in this case.